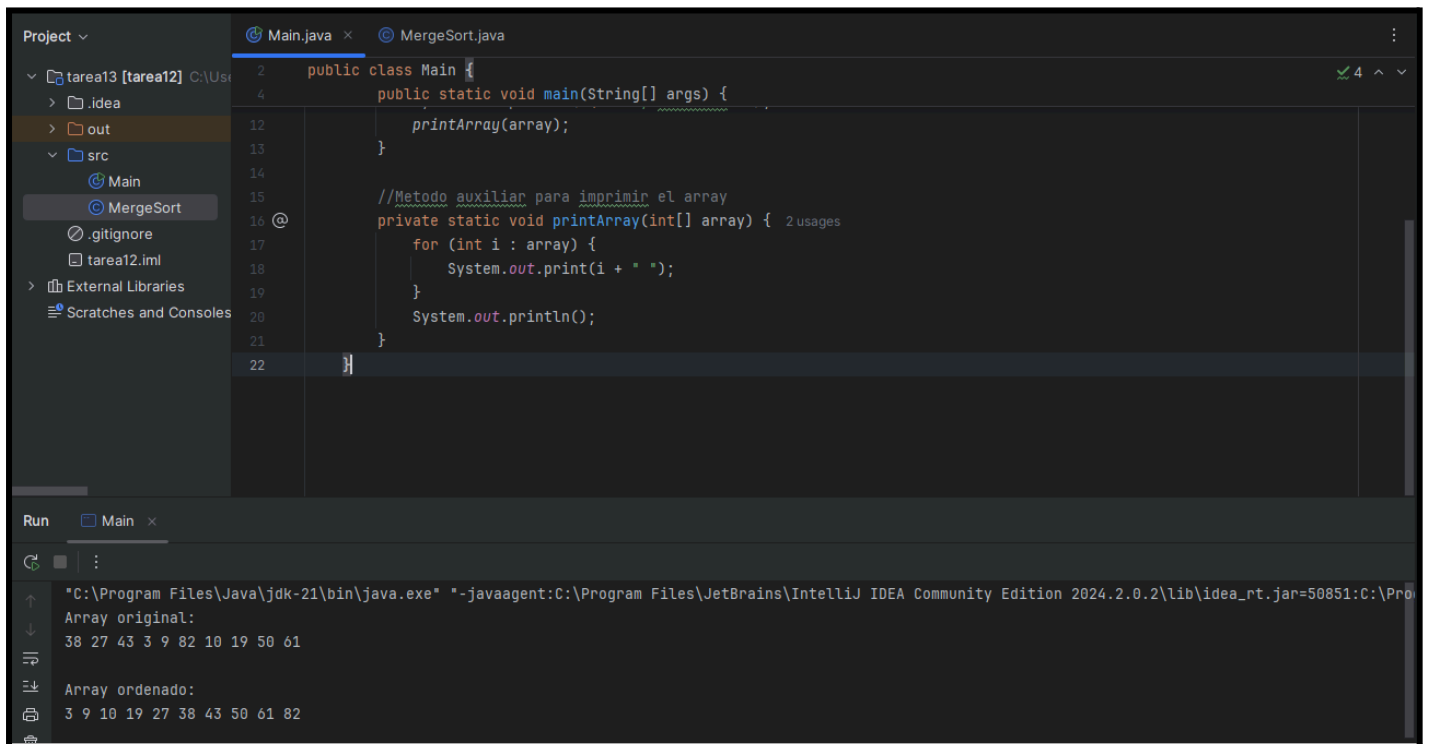


Escribir el algoritmo mergeSort en java.



The screenshot shows the IntelliJ IDEA interface with two files open: `Main.java` and `MergeSort.java`. The `MergeSort.java` file contains the implementation of the merge sort algorithm. The `Main.java` file contains the `main` method that initializes an array and calls the `mergeSort` method. The Run window at the bottom shows the output of the program.

```
2 public class Main {
4     public static void main(String[] args) {
12         printArray(array);
13     }
14
15     //Metodo auxiliar para imprimir el array
16     @ private static void printArray(int[] array) { 2 usages
17         for (int i : array) {
18             System.out.print(i + " ");
19         }
20         System.out.println();
21     }
22 }
```

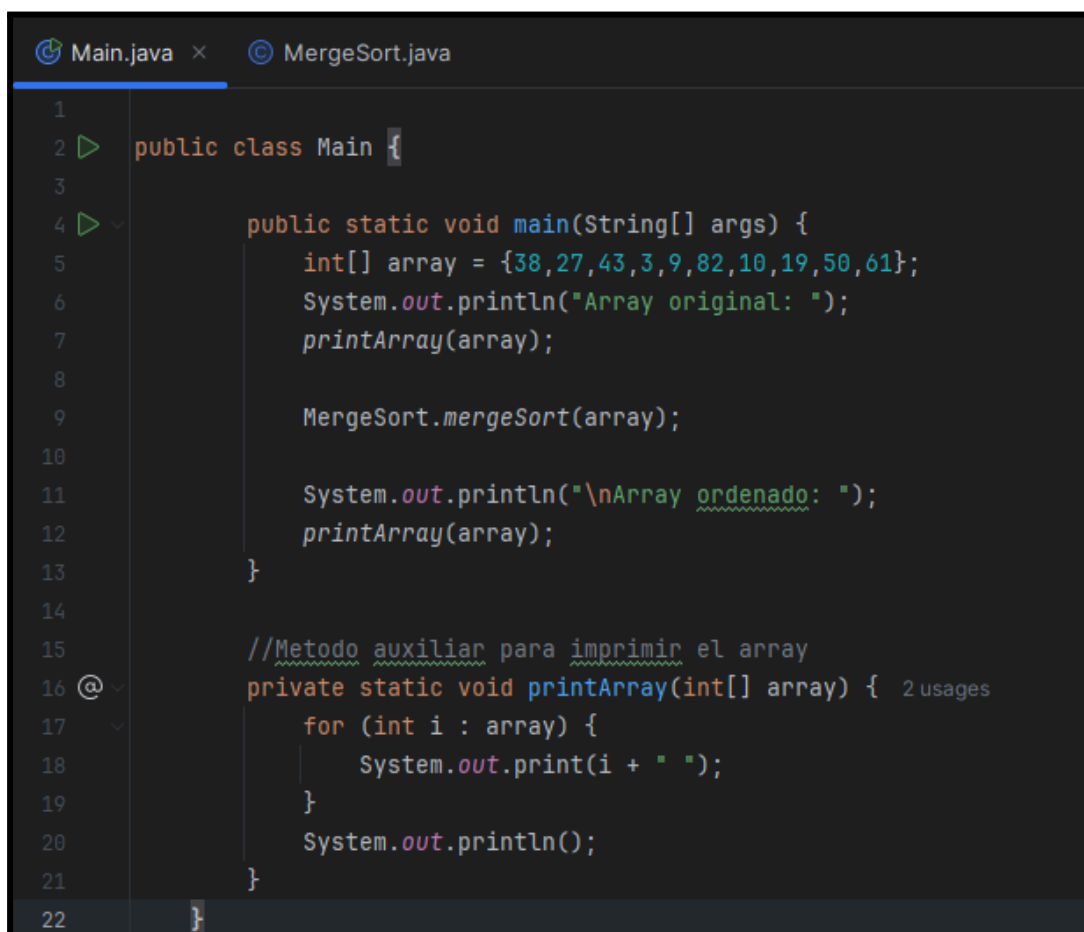
Run Main x

"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.2.0.2\lib\idea_rt.jar=50851:C:\Pro...

Array original:
38 27 43 3 9 82 10 19 50 61

Array ordenado:
3 9 10 19 27 38 43 50 61 82

Main



The screenshot shows the IntelliJ IDEA interface with two files open: `Main.java` and `MergeSort.java`. The `Main.java` file contains the `main` method that initializes an array and calls the `mergeSort` method. The `MergeSort.java` file contains the implementation of the merge sort algorithm.

```
1 public class Main {
2     public static void main(String[] args) {
3         int[] array = {38,27,43,3,9,82,10,19,50,61};
4         System.out.println("Array original: ");
5         printArray(array);
6
7         MergeSort.mergeSort(array);
8
9         System.out.println("\nArray ordenado: ");
10        printArray(array);
11    }
12
13    //Metodo auxiliar para imprimir el array
14    @ private static void printArray(int[] array) { 2 usages
15        for (int i : array) {
16            System.out.print(i + " ");
17        }
18        System.out.println();
19    }
20 }
```

Método MergeSort

```
1 public class MergeSort { 1 usage
2     //Metodo principal
3     @ public static void mergeSort(int[] array) { 1 usage
4         if (array.length <= 1) return;
5         int[] tempArray = new int[array.length];
6         mergeSort(array, tempArray, leftStart: 0, rightEnd: array.length - 1);
7     }
8
9     //Metodo recursivo para dividir el array y luego fusionarlo
10    private static void mergeSort(int[] array, int[] tempArray, int leftStart, int rightEnd) { 3 usages
11        if (leftStart >= rightEnd) return;
12
13        int middle = (leftStart + rightEnd) / 2;
14        mergeSort(array, tempArray, leftStart, middle);
15        mergeSort(array, tempArray, leftStart: middle + 1, rightEnd);
16        mergeHalves(array, tempArray, leftStart, rightEnd);
17    }
18
19    //Metodo para fusionar las mitades divididas del array
20    private static void mergeHalves(int[] array, int[] tempArray, int leftStart, int rightEnd) { 1 usage
21        int leftEnd = (rightEnd + leftStart) / 2;
22        int rightStart = leftEnd + 1;
23        int size = rightEnd - leftStart + 1;
24
25        int left = leftStart;
26        int right = rightStart;
27        int index = leftStart;
28    }
```

```
1 public class MergeSort { 1 usage
20    private static void mergeHalves(int[] array, int[] tempArray, int leftStart, int rightEnd) { 1 usage
28
29        //Fusiona los elementos de ambas mitades en el array temporal
30        while (left <= leftEnd && right <= rightEnd) {
31            if (array[left] <= array[right]) {
32                tempArray[index] = array[left];
33                left++;
34            } else {
35                tempArray[index] = array[right];
36                right++;
37            }
38            index++;
39        }
40
41        //Copia los elementos restantes de la mitad izquierda, si quedan
42        System.arraycopy(array, left, tempArray, index, length: leftEnd - left + 1);
43        //Copia los elementos restantes de la mitad derecha, si quedan
44        System.arraycopy(array, right, tempArray, index, length: rightEnd - right + 1);
45        //Copia el array temporal de vuelta al array original
46        System.arraycopy(tempArray, leftStart, array, leftStart, size);
47    }
48
49 }
```